

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

December 2012

EC 309 Digital Communication

Date: 06.11.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) In Random experiment, a trial consists of four successive tosses of coin. If we define an RV x as the number of heads appearing in a trial, determine $P_x(X)$ and $F_x(X)$. [04]
- (b) Define Aliasing. [03]
- Q - 2 (a) Explain the operation of Delta Modulation using neat block diagram and waveforms. [06]
What are the disadvantages of Delta modulation? How can we overcome them?
- (b) Explain Conditional Probability with example and derive Bay's rule. [06]
- (c) What are the desirable properties of line coding? [02]

OR

- Q - 2. (a) Explain the technique of Companding in PCM? State law for the same. [06]
- (b) State and explain the Central Limit Theorem. [06]
- (c) Define Regenerative Repeaters. [02]
- Q - 3 (a) Write a short on Optimum Binary Receiver. [06]
- (b) Draw the block diagram of digital communication system and explain the three major signal processing tasks. List the advantage of Digital Communication over Analog Communication. [08]

OR

- Q - 3 (a) Write a short note on coherent detection of Amplitude Shift Keying. [06]
- (b) Explain briefly Detection Probability. [08]

SECTION - II

- Q - 4 (a) A message source generates one of four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2 and 0.1. Each emitted message is independent of the other messages in the sequence. [04]
1. What is the source entropy?
 2. What is the rate of information generated by this source (in bits per second)?

- (b) Suppose we wish to increase reliability by repeating a message three times, e.g., by transmitting data 0 by 000 and 1 by 111. This is a (3, 1) code. [03]

1. Is this a systematic code?
2. If so, find the generating matrix G.

- Q - 5 (a) A zero-memory source emits six messages with probabilities 0.30, 0.25, 0.15, 0.12, 0.10 and 0.08 respectively. Find the 4-ary (quaternary) Huffman code. Determine its average word length, efficiency and redundancy of the code. [08]

- (b) Construct a systematic (7, 4) cyclic code using generator polynomial $g(x) = x^3 + x^2 + 1$. For data vector $d = 1010$. [06]

OR

- Q - 5 (a) Show that entropy is maximum when all messages are equiprobable. [08]

- (b) Explain Burst-Error detecting and Correcting Codes. [06]

- Q - 6 (a) Golay's (23, 12) codes are three-error correcting codes. Verify that $n=23$ and $k=12$ satisfies the Hamming bound exactly for $t=3$. [05]

- (b) Explain briefly QPSK modulation and BPSK modulation with neat sketch. [06]

- (c) A coin is tossed four times in succession. Determine the probability of obtaining exactly two heads. [03]

OR

- Q - 6 (a) A ternary code (11, 6) exists that can correct up to two errors verify that this code satisfies the hamming bound exactly. [05]

- (b) Write a short note on M-ary-FSK and M-ary-PSK. [06]

- (c) A binary symmetric channel error probability is P_e . The probability of transmitting 1 is Q , and that of transmitting 0 is $1 - Q$. determine the probabilities of receiving 1 and 0 at the receiver. [03]